

AI-ASSISTED CHRONIC CARE TRIAGE FOR OUTPATIENT DIABETES FOLLOW-UP

INDUSTRY CONTEXT AND PROBLEM DEFINITION

Healthcare systems face a recurring operational problem: many patients with chronic conditions show early signs of worsening risk before their next scheduled visit, but care teams do not always have enough time to review every chart in depth. In outpatient diabetes management, these early signals may include rising A1C levels, missed appointments, elevated blood pressure, lower medication adherence, and recent emergency room utilization. If these indicators are not recognized and acted on quickly, the result can be worse patient outcomes, avoidable acute care use, and higher system costs.

This problem is well suited to an AI-enabled workflow because it combines structured data analysis, predictive prioritization, communication support, and operational decision routing. At the same time, it is not a setting where full automation is appropriate. Healthcare is a high-stakes domain in which bias, privacy, explainability, and accountability matter as much as technical performance. The value of an industry-focused capstone solution, therefore, is not simply building a model; it is designing a coherent system that is technically plausible, ethically grounded, and professionally defensible.

The integrated solution I designed for this project is an **AI-assisted chronic care triage system** for outpatient diabetes follow-up. Its purpose is to help a clinic sort patients into actionable risk tiers, generate a short explanation of why a case is being prioritized, and route the case to the appropriate follow-up path. The system does not diagnose, prescribe, or replace a clinician. Instead, it supports earlier attention for patients who may need help while maintaining human oversight for high-risk outputs.

OVERVIEW OF THE INTEGRATED SOLUTION

The artifact combines four connected layers. First, a data-analysis layer profiles patient variables that are commonly associated with near-term care needs: age, A1C, systolic blood pressure, missed appointments, medication adherence, and prior ER visits. Second, a supervised machine learning layer estimates the probability that a patient needs intervention in the near future. Third, a generative layer translates the structured model output into a concise natural-language case summary and patient-facing outreach recommendation. Fourth, an agentic workflow layer routes the case to an automated reminder path, a nurse outreach path, or a

physician review path based on the predicted level of concern.

This architecture reflects the reality that useful AI systems in industry rarely depend on a single technique. Instead, they integrate analytics, prediction, explanation, and orchestration into a workflow that matches how people actually work. The design is intentionally modular so each layer can be evaluated separately while still contributing to a unified operational purpose.

INTEGRATION OF PRIOR PROJECTS AND METHODS

This project is explicitly built from methods and perspectives developed across earlier capstone work. The first influence came from prior data and statistical reasoning exercises, which trained me to identify relevant variables, summarize distributions, and reason carefully about patterns before modeling. That earlier work shaped the system’s front end: instead of sending raw data directly into a model, I structured the pipeline around interpretable clinical indicators and exploratory summaries.

The second influence came from prior machine learning work. Earlier predictive modeling projects emphasized the importance of feature selection, train-test separation, interpretable performance metrics, and the need to align modeling choices with the problem context. Those lessons directly informed my decision to use an interpretable logistic-style risk model rather than a more opaque architecture. In a healthcare triage setting, a model that can be explained and defended is more useful than one that is only marginally more accurate but difficult to justify.

The third influence came from prior generative AI activities. Those projects demonstrated how model outputs become more valuable when they are translated into readable, context-aware language for human decision-makers. In this capstone solution, the generative layer converts a numeric risk score into a plain-language summary of the key drivers behind a patient’s prioritization. That step matters because care teams need more than a score; they need a narrative they can act on.

The fourth influence came from prior agentic workflow design. Earlier work on tool chaining and multi-step orchestration showed that decision support becomes more useful when the system can recommend the next operational action rather than stopping at prediction. That insight informed the routing layer in this project. Once risk is estimated, the system decides whether the patient should receive an automated reminder, direct outreach from a nurse, or escalation

to a physician reviewer. In other words, the output of one layer becomes the input to the next, which is what makes the system integrated rather than merely multi-part.

TECHNICAL DESIGN DECISIONS AND TRADEOFFS

Several design decisions shaped the final artifact. The most important was the choice to use a synthetic dataset for demonstration. This decision reduced privacy concerns and made it possible to show the full pipeline without exposing real patient information. The tradeoff is that the prototype is technically illustrative rather than clinically validated. That is acceptable for a capstone synthesis artifact because the project’s emphasis is on system design, integration, and justification rather than production deployment.

A second design decision was model interpretability. I chose a logistic-style classifier because it supports a clear explanation of how structured variables influence risk. In a safety-sensitive domain, interpretability supports trust, stakeholder communication, and responsible governance. The tradeoff is that a more complex model might capture nonlinear relationships more effectively. However, for this use case, explainability and defendability were more valuable than maximizing complexity.

A third design decision was to keep the generative layer constrained. Instead of allowing open-ended generation, the system uses structured variables to produce targeted summaries and outreach drafts. This reduces hallucination risk and keeps outputs tied to observable patient indicators. The tradeoff is that the language is narrower and less flexible than a fully open generative model, but the gain in reliability is worth it.

A fourth design decision was to make the routing layer conservative. Critical and high-risk outputs trigger human review rather than autonomous action. This aligns with guidance from both NIST’s AI Risk Management Framework and WHO’s ethics guidance for health AI, both of which emphasize trustworthiness, accountability, and human-centered governance in high-impact systems (NIST, 2023; WHO, 2021).

ETHICAL, GOVERNANCE, AND RESPONSIBLE AI CONSIDERATIONS

The central ethical concern in this project is not whether AI can prioritize patients, but whether it can do so fairly, transparently, and safely. Healthcare data often reflect structural inequities in access, documentation, and prior treatment. If such patterns are learned uncritically, the system may reproduce or amplify unfair outcomes. Obermeyer et al.

(2019) showed that health management algorithms can encode racial bias when proxy variables such as historical cost are used without sufficient scrutiny. That finding is directly relevant here because triage systems can appear objective even when they are shaped by unequal historical conditions.

To address this risk, the prototype includes several safeguards at the design level. First, it uses a transparent set of clinical and behavioral indicators rather than obscure proxies. Second, it routes high-risk cases to human review instead of acting autonomously. Third, it is framed as decision support, not decision replacement. Fourth, it makes limitations explicit, including the need for subgroup fairness testing before any real deployment.

Transparency is another major ethical issue. A risk score without explanation can create automation bias, in which staff trust the output simply because it appears technical. The generative explanation layer is intended to reduce that problem by making the main drivers legible to clinicians and staff. That design choice aligns with broader calls for human-centered AI in medicine, where systems are most valuable when they enhance human judgment rather than displace it (Topol, 2019).

Privacy and governance also matter. Any real deployment would require secure handling of electronic health record data, role-based access controls, audit logging, and organizational review before use. WHO (2021) stresses that AI for health must place ethics and human rights at the center of design and governance. In practice, that means the system would need clear accountability, documentation of intended use, escalation policies, and ongoing monitoring after launch.

LIMITATIONS AND RISKS

The prototype has several important limitations. The most obvious is that the dataset is synthetic. This protects privacy during development but also means the results cannot be interpreted as clinical evidence. The model performance metrics show internal coherence within the simulation, not external validity in a real population.

Another limitation is scope. The current version focuses on a small group of structured variables and does not include lab history over time, social determinants of health, clinician notes, or medication changes. Those omitted factors could materially affect the quality of prioritization. There is also a risk of false negatives, in which a patient who needs intervention is labeled as low or moderate risk. In healthcare, that error can be more serious

than a false positive because it may delay needed follow-up.

A third limitation is workflow adoption. Even a technically sound model may fail if staff do not trust it, if alerts are too frequent, or if it adds administrative burden. That is why the system must be evaluated not only for predictive performance but also for usability, fit within the care process, and effect on staff workload.

PROFESSIONAL AND INDUSTRY RELEVANCE

This project demonstrates professional readiness because it goes beyond a narrow modeling task and addresses what employers and research stakeholders actually expect: cross-domain synthesis, defensible design tradeoffs, ethical awareness, and clear communication. In practice, AI professionals are often asked to connect analytics, modeling, user experience, governance, and operations in a single solution. That is precisely what this artifact attempts to show.

The project also reflects a realistic professional posture. Rather than claiming that AI should make clinical decisions on its own, the design positions AI as a structured support system inside a human workflow. That framing is more credible for industry use and more consistent with responsible deployment standards. It demonstrates not only technical fluency, but judgment.

FUTURE EXTENSIONS OR IMPROVEMENTS

If I were extending this work, the next step would be to validate the system on a properly governed real-world dataset with subgroup performance analysis. I would also expand the feature set to include longitudinal trends and social-risk indicators, while carefully checking for fairness implications. On the operational side, I would add monitoring dashboards, structured user feedback, and threshold tuning based on clinic capacity. Finally, I would formalize model documentation with a model card or similar governance artifact so that intended use, limitations, and known risks are visible to all stakeholders.

CONCLUSION

This capstone synthesis project demonstrates how a real-world healthcare problem can be addressed through the intentional integration of data analysis, predictive modeling, generative explanation, and agentic workflow design. More importantly, it shows that industry-

relevant AI work is not only about building systems that function, but also about building systems that can be explained, justified, governed, and defended. That combination of technical integration and responsible reasoning is what makes the project a meaningful capstone artifact.

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